

Day 6 — Bar Model Proof

Module: Subtraction — Nikhila (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: draw a visual proof using bar models and number lines.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pratiyogi* (modern coinage) (IAST: pratiyogi (modern coinage); lit. “the number that adds to a base”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Draw a bar of length 100.** Cut a section of length 37 from the right end. Label the remaining bar.
2. **Decompose using 99.** Show that the bar of 100 = bar of 99 + bar of 1. The bar of 99 is easy to subtract from (every digit is 9).
3. **Visual mapping to digits.**
 - Top row: 99 partitioned into a section of 37 and a section of $(99-37) = 62$.
 - Bottom row: add the unit 1, splitting it into the rightmost cell.
 - Result: $(99-37)+1 = 100-37 = 63$.
4. **Generalise visually.** Show the same bar-decomposition for $1000-437$. The leading 9s allow digit-wise subtraction without borrow; the +1 at the end bumps the last digit.
5. **Independent task.** Each student draws their own bar-model proof for $1000 - 248 = 752$ and submits as proof of understanding.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Bar Model Proof.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).